

Athira Pillai

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EDUCATION

Master of Science (MS), Computer Science, University of Texas, Arlington (GPA: 3.75/4)

Aug 2023 – May 2025

Courses: *Design & Analysis of Algorithms, Advanced Software Engineering, Distributed Systems*

Bachelor of Engineering (BE), Computer Engineering, University of Mumbai (GPA: 8.3/10)

Aug 2016 – May 2020

Courses: *Database Management Systems, Machine Learning, Object Oriented Programming, Data structures*

PROFESSIONAL EXPERIENCE

Volunteer Research Assistant - *University of Texas at Arlington (Texas, USA)*

July 2025 – Aug 2025

- Designed and implemented an LLM-based Q&A system leveraging prompt engineering with LangChain, Gemini, and FAISS, enabling accurate, real-time document summarization and semantic search.
- Developed a scalable ingestion pipeline in Python (merging, chunking, embedding, vector indexing), reducing manual data lookup time by 60% and supporting AI agents for strategic planning.

Software Developer - *Tata Consultancy Services PVT LTD (Mumbai, India)*

Feb 2022 – Oct 2023

- Led development of modular Azure DevOps automation tools in Python, orchestrating repo setup, dashboard creation, pipeline templates, and wikis via CI stages — reduced onboarding time by 80% across internal teams.
- Implemented OAuth 2.0 authentication with Azure service principals and Azure AD in CI/CD pipelines, eliminating token errors and improving pipeline security and reliability by 35%.
- Integrated SonarQube and Pytest into CI/CD pipelines for test coverage and quality gates, reducing bugs by 25% and enabling full validation before deployment.
- Engineered reusable Jira-Azure DevOps integration scripts to automate issue types and fields at the project template level, cutting manual setup by 60% and improving consistency across projects.
- Boosted team efficiency by 20% by mentoring junior developers in python and powershell, integrating agile methodologies and optimizing code reviews to ensure adherence to best practices.

Assistant Software Developer - *Tata Consultancy Services PVT LTD (Mumbai, India)*

Feb 2021 – Feb 2022

- Developed a multithreaded document-processing tool in Python (Tkinter, Regex, OOP) to automate Azure wiki documentation and formatting, improving internal delivery efficiency by 60%.
- Built modular RESTful APIs with Flask for document processing, improving system maintainability and enabling seamless integration between front-end and back-end components.
- Automated detection of broken links using PowerShell and serverless Logic Apps workflows with alert instrumentation, decreasing system errors by 25% and improving monitoring reliability.
- Improved credential lifecycle management by integrating Azure Key Vault and automated PAT renewal workflows, reducing token-related errors by 90%.

PROJECTS

Smart Expense: Finance Tracker App (*Python Flask, SQL, HTML/CSS*)

- Built a full-stack finance tracker with Flask and SQLite, implementing responsive and interactive UI features with real-time categorized insights that reduced overspending by 30% via dashboard analytics. [\[GitHub Project Repo\]](#)
- Helped users reduce overspending by up to 30% through real-time dashboards and visualization of expense data.

Patient Monitoring System (*IoT and Cloud*)

- Engineered a NodeMCU-based health monitoring system integrated with ThingSpeak, enabling real-time tracking of patient vitals and automated emergency alerts, which improved response time to critical health events by 40% in test scenarios.

Emoticon Suggestion System (*Python, NLP, AJAX*)

- Engineered a real-time emoji recommender using NLP models and Levenshtein distance delivering 70 percent accuracy in dynamic suggestions through Flask and AJAX integration.

SKILLS

- Languages:** Python, C/C++, C#, Java, JavaScript, HTML, CSS, PowerShell, React.js, TypeScript
- Frameworks & Tools:** Azure DevOps, Postman, Git, Microsoft Azure, Figma, Flask, CI/CD, Responsive Design
- Cloud & DevOps:** Azure Cloud, AWS, CI/CD Tools, Agile Methodologies, Kanban boards, Jira, Docker, Kubernetes
- Databases:** SQL Server, MySQL, PostgreSQL, RDBMS, SQLite, NoSQL, Document Databases
- Certifications:** Microsoft Azure AZ-900, Microsoft Azure AZ-204
- Publications:** "Emoticon Suggestion with Word Prediction using NLP" — IRJET, Vol. 7 Issue 05. [\[Publication PDF\]](#)
- Programming Concepts:** Object-Oriented Programming, Software Architecture, Design Patterns, Software Testing, Full-Stack Development, Scalability, Performance Optimization, Observability, Automation Workflows, Event-Driven Architecture